

Day 1 Worksheet: The Tissue Systems & Vascular Anatomy

Epidermal, Ground & Vascular Tissue Systems | Open vs Closed Bundles | Stomatal Complex

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Student Name: Date: Time Allowed: 30 Mins Max Marks: 20 Marks Obtained:

Day 1 Focus Review: • Epidermal system = Epidermal cells + Stomata (guard cells + subsidiary cells) + Appendages (Trichomes/Root hairs). • Ground system = All tissues except epidermis & vascular bundles (Cortex, Endodermis, Pericycle, Pith, Mesophyll). • Vascular bundles: *Radial* (in roots; alternate radii) vs *Conjoint* (in stem/leaf; same radius). • *Open* (cambium present) vs *Closed* (cambium absent). *Exarch* (roots) vs *Endarch* (stems).

SECTION A: OBJECTIVE & MULTIPLE CHOICE QUESTIONS (1 MARK EACH)

1. Dumb-bell shaped guard cells are characteristically found in:

1M

- (a) Mango & Sunflower (b) Grasses & Wheat (c) Mustard & Pea (d) Ferns & Mosses

2. Vascular bundles in which cambium is present between xylem and phloem are called:

1M

- (a) Closed (b) Radial (c) Open (d) Exarch

3. Multicellular epidermal hairs present on the stem are known as:

1M

- (a) Root hairs (b) Trichomes (c) Bulliform cells (d) Guard cells

4. Ground tissue of a green foliage leaf is represented by:

1M

- (a) Stele (b) Mesophyll (c) Pericycle (d) Endodermis

SECTION B: SHORT ANSWER & CONCEPTUAL QUESTIONS (2 MARKS EACH)

5. Differentiate between Trichomes and Root Hairs (Give 2 distinct points).

2M

6. Distinguish between Exarch and Endarch xylem conditions. Mention where each is found.

2M

7. Differentiate between Radial and Conjoint vascular bundles with a diagrammatic sketch.

2M

SECTION C: DESCRIPTIVE & BOARD-LEVEL QUESTIONS (3 & 4 MARKS)

8. What is Stomatal Apparatus? Explain its three anatomical components in detail.

3M

9. Classify vascular bundles based on (a) Radial arrangement and (b) Presence or absence of cambium. Explain the significance of open vascular bundles.

4M